Sample Document using `adenc.sty`

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Check out the Github Repository for `adenc.sty`!

Engine requirement: `adenc.sty` requires XeLaTeX or LuaLaTeX. pdfLaTeX is not supported (`fontspec` and `unicode-math` are engine-specific).

1 Package Options

Options may be passed to the package in the usual way.

Option	Values	Description
`theorembox`	`none*`, `plain`, `color`	Controls the theorem environment style. `plain` adds `tcolorbox` boxes; `color` adds colored boxes. See Section 2.
`hideproofs`	`false*`, `true`	Hides proof environments.
`hidemarkings`	`false*`, `true`	Hides markings generated by `\markabove` and `\markbelow` (see Section 3).
`workingpaper`	`false*`, `true`	Adds a watermark with the date on the first page and expands the margin so notes created with `\todo` can fully display.
`header`	`true*`, `false`	Controls whether the running header is displayed.
`clearsection`	`true*`, `false`	Controls whether sections start on a new page. When on, header will not be displayed on first page of each section.
`fancyqed`	`true*`, `false`	Controls whether environments such as `remark` and `example` end with a QED-style symbol.
`pset`	`false*`, `true`	Preset for problem sets: sets `clearsection=true`, `header=false`, and `tocdepth` to 1. Colors `problem/exercise/question` dark purple in `theorembox=none` mode. Options listed after `pset` override its defaults.
`index`	`false*`, `true`	Auto-index every `\vocab` term and print the index at end of document.

* default value

2 Theorem Environments

Definition 2.1. A definitive **definition** is a definition, by definition.

Lemma 2.2. *A lamentable lemma.*

Theorem 2.3. *A towering theorem.*

Corollary 2.4. *A cool corollary.*

Remark 2.5. A remarkable remark. Note that I intentionally keep the remark environment plain.

Example 2.6. An exemplary example.

Problem 2.7. A problematic problem.

Algorithm 2.8. A well-specified algorithm.

Proof. A precise proof. □

Proposition 2.9 (A long proposition that breaks across pages). *Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.*

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penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.
Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.
Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Numbering can be turned off by using the corresponding `*` versions of the environments (e.g. `theorem*` instead of `theorem`).

3 Marking the Document

The `\todo` command in the `todonotes` package is a great way to add notes to a document, but among other things, it does not support display style math and, when used frequently, the places to which they point can be hard to decipher. It is for these reasons that the following commands are introduced:

E.g., this points to the word "can."

And this to "hard."

(a) `\begin{itodo} ... \end{itodo}` (inline todo) produces an inline block of notes. This can be used as placeholders for contents to be added later.

This is an example of what notes produced by the `itodo` environment looks like. Unlike the `\todo` command, the `itodo` environment supports display math:

$$\sum_{n=1}^{\infty} a_n z^n.$$

(b) `\markabove` and `\markbelow` provide a way to mark texts without altering the spacing. Both commands take two arguments: (1) align method (`l`, `c`, or `r`); and (2) text to display. For example,

test1
Test test test test
I'm marking below here
Test test test test.
above!
Test test test test
math! α

is produced by the following code:

(c) `\draftnote[Label][color]{text}` produces a short inline draft comment. The label and color are optional; they default to `Note` and red, respectively. For example, `\draftnote{fix this}` produces [Note: fix this] and `\draftnote[TODO][blue]{rewrite}` produces [TODO: rewrite].

(d) `\mnote[offset]{text}` places a boxed margin note at the current vertical position. The optional offset shifts the note vertically, e.g. `\mnote[-1ex]{note}`.

This is a margin note.

```
Test test test\markabove{l}{test1} test\markbelow{c}{I'm marking below here}.

Test test test test.

Test test test\markabove{c}{above!} test\markbelow{r}{math! \(\alpha\)}.
```

4 New Commands

Some commands (mainly for math symbols) are added or modified for aesthetics and/or convenience. A few notable ones are mentioned below:

Description	Example	LaTeX Commands
Command for styling new vocabulary[1]	**word**	`\vocab {word}`
Contradiction symbol[2]	※	`\contradiction`
Shortcuts for `\mathbb`	$\mathbb{R}, \mathbb{Q}, \mathbb{F}, \mathbb{P}$	`\bR , \bQ , \bF , \bP`
Shortcuts for `\mathcal`	$\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$	`\cA , \cB , \cC , \cD`
Shortcuts for `\mathsf`	$\mathcal{L}, \mathcal{T}, \mathcal{U}, \mathcal{V}$	`\sL , \sT , \sU , \sV`
Better looking complement symbol	$A^\complement$	`A^\complement`
Better looking empty set symbol	$\emptyset$	`\emptyset`
Command for vectors	$\mathbf{v}$	`\vec {v}`
Shortcuts for matrices	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	`\bmat {1 & 2 \\ 3 & 4}`
	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	`\pmat {1 & 2 \\ 3 & 4}`
	$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$	`\vmat {1 & 2 \\ 3 & 4}`
Differentiation operators	$\mathrm{D}f$	`\DD f`
	$\int f \,\mathrm{d}x$	`\int f \dd x`
Imaginary number	i	`\ii`
The indicator function	$\mathbb{1}$	`\ind`
Independent	$\perp\!\!\!\perp$	`\indep`

[1] In, for example, definitions.
[2] Andreas Stavrou taught me this symbol.

5 Document Commands

`\courseinfo \{code\}\{term\}\{title\}\{lecturer\}\{notetaker\}` sets the document title and author for course lecture notes. For example:

```
\courseinfo{ECMA308}{W26}{Theory of Auctions}{Philip Reny}{Aden Chen}
```

produces the equivalent of:

```
\title{ECMA308 (W26): Theory of Auctions}
\author{{Lecturer: Philip Reny} \\ {Notes by: Aden Chen}}
```

`\makepsethead [due date]\{course\}\{assignment\}\{author\}` renders a centered header block (course name, assignment title, author, compile date) followed by the table of contents. Call it at the top of `\begin{document}`. The optional due date is omitted when not provided.

```
\makepsethead[2026-04-30]{MATH 101}{Problem Set 3}{John Smith}
```

`\iproblem [label]\{text\}` typesets an inline problem statement in dark purple with a trailing QED-like symbol. The optional label (e.g. a problem number) is rendered in bold. Intended for use with the `pset` option.
1. Show that $\mathbb{R}$ is uncountable.
Produced by:

```
\iproblem[1.]{Show that \(\RR\) is uncountable.}
```

See `pset.tex` for a complete sample problem set using the `pset` option.

A Credits

The first version of `adenc.sty` drew inspiration from:

- Andrew Lin's package `lindrew.sty`.
- Gilles Castel's lecture notes repository.
- A TeX StackExchange discussion on robust draft text marking.
- A Math StackExchange discussion on symbols for contradiction.